

CitySense: blockchain-oriented Smart Cities

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ABSTRACT

A smart city is a connected system in which things produce a huge quantity of data. We focused the attention on monitoring the environment quality in urban area by means of a distributed network of small mobile sensors that are devices on the Internet of Things (IoT). Sensors produce digital measurements, useful for investigating and studying the life quality in every part of the city. In our vision, environmental data must be available by everyone and shared with citizens, but it must be unmodifiable. We propose to solve the problem of the sensors data storage and management using a disruptive technology called blockchain. The blockchain responds to the demand of availability and unchangeability and, thanks to the potentiality of smart contracts, makes us able to manage sensor information and implement a control logic. In order to develop the software based on blockchain we chose to apply the SCRUM methodology because of its capabilities of being a flexible, adaptive and iterative methodology.

KEYWORDS

Blockchain, Smart cities, Internet of Things, Sensing, SCRUM

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1 INTRODUCTION

What is a Smart City? It is an urban system where some requirements exist simultaneously: a functional management model of traffic and public transport, a context in which citizens can work remotely, without moving, in a lot of activities, the opportunity to live in safety thanks to the use of innovative solutions of public

surveillance, assistance and the application of appropriate technology for energy saving and to reduce the environmental impact. Linked to these technical aspects we want to identify some political and human issues which are equally important. In fact, a Smart City is able to create new ways of participation in public life and calls on people to become active citizens in order to enhance a continuous and two-way relationship with local government [15].

In this complex scenario the implementation of an urban IoT may be an excellent solution for real optimization and management of public services. Several kinds of data, collected by sensors in a IoT system [6], can be used to increase the openness of public government and political choices, to improve the people's awareness about well-being of the city and to encourage the involvement of citizens in the drive for sustainable development. In order to achieve these goals, we have chosen blockchain technology as public available shared database and then as instrument of reliability, transparency and security. Moreover, we are going to apply SCRUM methodology to build this system blockchain based and we are going to involve citizens in the software development. In this way, we can make a real open source community.

The paper is structured as follows. Section 2 contains a brief discussion of related works. Section 3 presents our approach, describing the typology of sensors we use and the way we manage information by means the blockchain technology. We proposed a discussion in section 4 and finally section 5 ends the paper providing conclusions.

2 RELATED WORKS

Smart Cities projects are focused on viability, digital services, payments and environment, but it should be improved in the following key areas: provide better ground coverage of the services, as every item can be both client and server, and distribute the processing tasks (for power, money, time, etc.) among a big number of calculators capable of simple operations. Modern smartphones are up, in performance, to some of the mainframe servers deployed 20 years ago in the ISP industry [8]. Tailored software and simple computation tasks can be written to be executed by such small devices with good performances. Across Europe, several cities have been engaged in environment safeguard plans, starting from the

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2008 SETIS Plan to reduce CO₂ emission that widespread over 12 countries and 200 mid-sized towns, up to single town's projects to reduce power consumption and invest on new energy markets while the industrial ones that relies on technologies and sources from the last century are slowly fading away.

Emphasizing on the social approach of a smart city plan would be the key for raising the interest of the citizens who will become the main actors in this project, for contributing to the mass effect of self knowledge of the environment they live in and for keeping it as wealthy as possible. This social approach can be interpreted correctly by the blockchain technology. In fact, after its introduction, the blockchain technology has interested more and more scientists who study potentialities and applications in a multidisciplinary world of topics. For instance, researchers studied aspects related to the network [10], economy [13], and software engineering [11]. We reported the most recent publications, which show the state of art in the field of blockchain and in smart contracts applications to smart cities and Internet of Things. The feasibility of a blockchain-oriented smart city is an in-progress study. Looking in the web, is easy to find related initiatives, like that of the Dubai government which is programming to create a blockchain based smart city ¹, and debates about potentiality of this application ². Sharma et al. [12] studied an application of blockchain technology to build a vehicle network which takes in account several problematics that are the mobility of nodes (which represent vehicles), the confidentiality and security of data. Security aspects are also discussed by Biswas et al. in their work [1]. We take this work into particular consideration since it work discusses the need of a specific security framework.

Such framework is composed of four layers: the physical layer (which includes sensors), the communication layer (in which is considered the blockchain), the database layer and the interface layer (that considers all applications). Considering the IoT a key element for the smart cities development, we briefly discuss IoT related applications of the blockchain technology. In fact, it is an enabling technology for empowering the potentiality of the IoT. The work of Quaddah et al. [9] provides a well-defined framework named FairAccess to enable the communication between nodes by means of some blockchain based mechanisms (i.e smart contracts and transactions). Thanks to the blockchain, this framework provides a stronger and transparent access control tool. E-business aspects of IoT technology are discussed by Zhang [16]. He studied a blockchain application which he implements a seller-buyer model describing business operations between two or more devices. Christidis and Devetsikiotis [2] provide a discussion based on the literature, proving that smart contracts and blockchain applied to the IoT can be pretty powerful. They also provide an interesting section about the blockchain taxonomy. IoT (RFID based) and blockchain can also enable products traceability in a supply chain, as discussed by Tian [14], and can also enable the control of remote robots, as discussed by Ferrer [5]. Recent studies [7] have shown that a smart city needs to have a smart local council and a smart methodology in order to develop an efficient software. A good way to achieve this

goal is the use of SCRUM process with some changes with respect to the original approach.

3 THE PROPOSAL

Networks of Things acquire and share information on what is happening of interest to the citizens and can be used within the city to build a smart system of services.

The Smart Object network will be able to receive and store information from the surrounding environment through the use of sensors, in order to measure the values of the phenomena of interest, and through the use of actuators that act on the context. Our idea is to develop an IoT application, based on blockchain, to investigate some aspects of urban setting, to acquire relevant data and to process the information received.

We propose to use SCRUM methodology in order to implement this innovative blockchain-based system. We chose to apply a SCRUM process because of its capabilities of being flexible, adaptive and iterative. These aspects are perfectly in line with the intrinsic properties of blockchain, like decentralisation and reliability, and especially like the capability of managing data, transactions and all changes in a much faster and more efficient way. Therefore, we believe that Scrum may constitute a suitable methodology in order to increase the communication between users who send useful data recorded on the blockchain, and in order to exploit the continuous evolution of software and the collective accountability.

According to SCRUM methodology, in order to verify the correctness of system developed, the client's role is extremely important in several aspects at all the stages of the process. The contribution of customers impacts directly on product quality. For this reason, the clients must be involved in any decision adopted and shall collaborate closely with the development team. In our work, we consider as clients all citizens who wish to contribute to well-being of the city through the transmission of data from mobile devices by means a specific application. The same application will allow the transfer of information that comes from sensors. It will also be able to provide other functionalities, like sending messages, providing reports and user feedbacks. The data can be recorded on blockchain by means of a connection to the peer-to-peer network. The Product Owner will be responsible for analysing information provided by users through the mobile applications, for interpreting consumer expectations and requirements, for filtering communications transmitted, for identifying priorities and for distributing tasks within the development team. The local government will be involved by the Product Owner in the analysis of information communicated by users in order to provide feedback to the developers and take decisions. The city council then, through its decisions, shall participate in all phases of work and for each sprint from preparation of the public tender to monitoring of activities performed. This method favours effective application of an entirely cooperative approach.

¹<http://www.coindesk.com/dubais-museum-future-sees-blockchain-smart-cities/>

²<https://dcebrief.com/blockchain-powers-new-smart-city-initiative/>

3.1 The CitySense system

In order to conduct our study, we have identified some measurable phenomena and their characterizing quantities (noise, concentration of hydrogen, methane, carbon monoxide and microparticulates in the air, temperature, humidity and light). These quantities are related to environmental pollution issues in which the public opinion is most involved because they have a direct impact on people's lives.

The increasing diffusion of small programmable embedded devices that are easily connected to networks via wireless technologies had increased the number of actors eligible to participate to IoT networks that can share information. From this perspective, peculiar to the IoT world, the creation of an ecosystem of small interconnected "Smart Objects", that can be considered as an extension to the human body (nowadays we think about smartphone, smartwatches, wristbands and portable gadgets) would bring to life a complex and intelligent network of distributed systems that can interfere, both in a good and in a bad way, with the surrounding environment.

If we apply this schema to a geographic area, such as a town or a county, we can imagine a full coverage of the ground made by portable devices that collect data and send them back to a central collection endpoint.

The data collection mechanism works by combining the measurements acquired by the mobile devices with the validation process made with a specific mining software that process the data.

Mobile phones are the best devices to be used for this task as they provide a quite good computational capability, moderate battery life, wireless connectivity to other near devices, internet access and they are easily programmable to run generic purpose software on top. To connect to hardware-level machine, like electronic devices and sensors, a HW/SW connection layer should be deployed. As Today, a lot of such programmable tiny operational boards equipped with AVR processing units could be used for this task. Miniaturized boards with small footprint up to 30mmx20mm [15] can be connected to small power sensors such as DTH-22 for temperature and humidity, TEMT6000 for light, GP2Y1010AU0F for PM8 detection, due to their low current requirements and MUXable digital I/Os, as the number of analogic inputs is low on this miniaturized boards. Bigger sensors, such as CO, Hydrogen, Methane, require a huge amount (1 Amp x 5/9 Volts) of current in comparison to the previous digital ones, as they need to heat a filament (just like happened in vacuum tubes) to be able to operate.

This current consumption has an impact also on portability of the complete kit, as the power requirements would be too much for the desired target (people on the move). On the other hand, employing small sensors like the one mentioned above, in conjunction with a BLE (Low Power Bluetooth 4.0 connectivity) can open up a wide variety of solutions, like the possibility to attach such device to a small solar panel, the size of a pocket mirror, or to be powered from a dynamo connected to a bicycle wheel. Also, the number of sensors deployed could be larger as the software elaborates the results in a more sophisticated way. Computational power on this task is not a real big issue as digital multiplexing on data is quite

CitySense: Layers

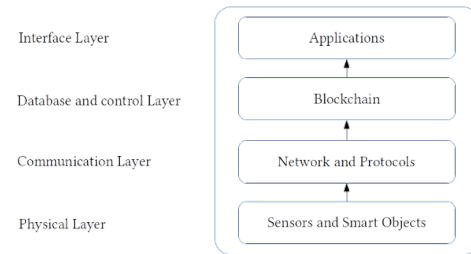


Figure 1

easy for this scenario. All the sensor package can be powered via a very common and cheap 3.7V LiPo battery, and even be powered from the USB OTG phone (if capable), that can boost up to 5V to 450 mA.

3.2 The blockchain solution

Blockchain is a decentralised and public available shared database, reachable by accessing to a peer-to-peer network. The blockchain is an enabling technology which allows to obtain the satisfaction of security and availability of data, and provides the computational power that makes it able to control the communication between nodes.

A blockchain contains a list of linked blocks. Information stored in blocks characterize the state of the blockchain. The state can change during the time, by means of the execution of the state transition function. The state transition function takes in input the previous state and a transaction. Each transaction must respect the validity conditions. Valid transactions are grouped and written in a new block by means of an operation called mining. As a result, each block inside the blockchain contains a list of validated transactions. Transactions can be seen as a way to send messages between two nodes, in a communication system. In a blockchain, each node of the communication system is called account and is identified by a fixed length hexadecimal number named address. Considering the Ethereum blockchain, the state is composed by all variables recorded in the blockchain and their values. These variables include all addresses balance and contracts internal variables. A contract is a special typology of account. It is recorded in a block of the blockchain, and can receive and transmit messages from and to other account, by means of transactions. Messages can request the execution of a specific contract functions. For instance, a message can contain the address of the receiver, the name of the function, and a list of parameters.

In order to develop the CitySense system, we plan to use the Ethereum platform to record measurements arriving from the IoT network of sensors. In our system, sensors are IoT devices, programmed to be connected with the blockchain and able to send messages. As in [1], CitySense is structured by layers as is shown in Fig. 1.

Layers of the system will be implemented in parallel. Each layer has specific and distinctive requirements and will be implemented

in an iterative process, with the advantages of the SCRUM methodology. The set of sensors composes the physical layer. In our system, sensors are carried around the city by their owners. For this reason, each measurement must be associated with the geographical position, in addition to the typology of measurement and to the timestamp. Considering the network layer, in our system sensors are programmed to send measurements to the blockchain through the peer-to-peer network by means of a light version of existent clients such as *geth*. Actually, sensors can be connected to a mobile device (for instance a smartphone) to run, or can be autonomous systems able to connect to the peer-to-peer network. Different communication mechanisms, such as WiFi, 3G or Bluetooth, can be used. Specific communication protocols should be defined considering the cost of a transaction in the Ethereum system (which is proportional with the payload size of the message), and designed according to a convenient data format.

In CitySense the blockchain consists in the database layer in which measurements are organized and stored. In our system we use contracts to control and save data. A first contract is designed to be the receiver of messages coming from sensors. We call it acquisition and sorting contract (ASC). Depending on the geographical position, the ASC sends the measurement to one of the set of specialized contracts that we call geographic contracts (GC). Inside each GC only measurements coming from a specific geographic area are stored. With specific implemented functions, GCs can be queried to provide measurements organized in several ways (for instance measurement can be organized by their typology, by timestamp, by value).

The variety of applications enabled by the CitySense system compose the application layer. Thanks to the collaborative nature of our system, the same people who participate to the creation of the CitySense are also able to take advantage of the information that is made available. For instance, a CitySense mobile application could be useful to people who want to know the environment quality of a specific area in the city. Furthermore, web applications could be used by a public administration to check if and where the city has pollution problems.

4 DISCUSSION

In our vision, blockchain represents the disruptive technology which will drive the development of future smart-cities related applications. Key features, such as to be a shared, transparent, distributed, secure, available and smart technology, make the blockchain an opportunity to improve potentialities of IoT and smart-cities development. As it is known, blockchain data are publicly available. In a smart-city, transparency makes citizen aware and able to know the contribution of each of them and how public governments use data. In case of need (i.s. sensitive or personal data) it is always possible to encrypt data before they are stored inside the blockchain.

In a blockchain based service, nodes participate to the objective in a democratic way, under the constraint of the consensus rule. Therefore, blockchain enables trust-free transactions without the need of a central control authority. IoT devices participate in

the peer-to-peer network sending messages to it, as nodes of the system and under the blockchain rules. In a blockchain system, smart contracts, computer programs located and working in the blockchain, can automatically acquire data from IoT devices and produce computed outputs. Because of the distributed nature, smart contracts cannot be modified or interrupted. For this reason, their usage could improve the reliability of a smart-city system.

IoT, SCRUM methodology and blockchain form the basis of digital transformation. A trust distributed technology ensure privacy, scalability, transparency and reliability. In a Smart City the number of linked objects is set to increase and will produce higher and higher operating costs. The blockchain will be a key element for the cost-cutting in the tracking and coordination of physical devices and will solve failure problems of traditional networks. Currently all IoT systems depend on client and server communication protocols, such as SSL and TLS and on cryptographic mechanisms such as the Public Key system, used in order to make the communication system verifiable and to authenticate the nodes of the network. Such architecture will soon have to face several problems (delay transmission for example) caused by data traffic congestion. Therefore, the decentralization, which is an intrinsic property of blockchain, could be the right solution in order to increase the network efficiency and reduce management costs. CitySense will allow a direct communication between smart devices and will verify the transactions without a central server.

Scrum methodology will be also used in order to ensure software quality, reduce the time-to-market, enhance the support of the citizens and create an infrastructure that allows the transfer of data in real time using a sensor networks with a low energy consumption [3]. The support of the citizens is essential to make this project succeed, but at the same time implies more complexity. For this reason we estimated a duration of 30 days for every sprint: this choice improves the collaboration of stakeholders. We planned also unitary tests for each iteration [4].

5 CONCLUSIONS

In this work we propose a blockchain based system to manage the environmental context. CitySense will be a system that encourages the constructive involvement of people in monitoring environment quality within the urban area of the city to promote a greater awareness of city health. We would like to promote the collective involvement of all citizens at all stages of software development. In order to do this we have chosen to use a SCRUM methodology in which the users' contribution is essential and the Product Owner must take into account what are citizens needs and their suggestions. In our approach the citizens' role is really important also in the development of software: they transfer information to the product owner, send him their feedback and participate in meetings at the end of every sprint.

The environmental data collection mechanism works by combining the measurements acquired by the mobile devices with the validation process made with a specific mining software that process the data. Local government, with little, or no monetary investments

at all, might give to all its citizens a digital device capable of acquiring data of the surrounding environment and create a health-map of the city they live in, in order to acquire real time data and prepare real time countermeasures to pollution alarms.

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